

Solving compound inequalities



1 Consider the statement $3 < 6x + 7 < 9$:

- a Write the two inequalities that make up $3 < 6x + 7 < 9$?
- b Identify the word joining the two inequalities that were obtained from the compound inequality $3 < 6x + 7 < 9$.

2 A high level of LDL cholesterol (“bad cholesterol”) in the blood increases a person’s risk of heart disease. The table shows how LDL levels affect risk.

Use x as the variable that represents the LDL cholesterol number to write a single or compound inequality for:

- a The optimal risk category
- b The Near optimal / above optimal risk category.
- c The Very high risk category.

LDL Cholesterol	Risk Category
Less than 100	Optimal
100 – 129	Near optimal / above optimal
130 – 159	Borderline high
160 – 189	High
190 and above	Very high

3 Determine whether the following statements are true for the compound inequality:
 $x - 2 < 4$ and $4x > 11$

- a Both $x = 3$ and $x = 6$ are solutions to the compound inequality.
- b Neither $x = 3$ nor $x = 6$ are solutions to the compound inequality.
- c $x = 3$ is a solution to the compound inequality.
- d $x = 6$ is a solution to the compound inequality.

4 Determine whether the following values of x are solutions to the compound inequality:
 $3 - x > -7$ and $3 - x \leq 6$

- a $x = -4$
- b $x = 0$



- 5 The steps to solve $2 < 6x + 8 < 10$ have been jumbled up below. Number the steps from 1 to 5.

Step Number	Step
⋮	Divide each part by 6.
⋮	Simplify by combining constant terms.
⋮	Subtract 8 from all three parts.
⋮	Simplify any divisions.
1	Write the given compound inequality.

- 6 The formula for converting temperatures from Fahrenheit to Celsius: $C = \frac{5}{9}(F - 32)$. During a recent year, the temperatures in London, ranged from -5°C to 35°C . Form a compound inequality to solve for the corresponding range of values of F , the temperature in London, in degrees Fahrenheit.
- 7 The velocity, in feet per second, of a tennis ball t seconds after it is projected directly upward is $v = 89 - 32t$. Solve for the range of times t at which the velocity will be between 41 ft/sec and 73 ft/sec (not inclusive). Write your answer as an inequality.
- 8 To get a final grade of B , a student must have an average mark on five tests that is greater than or equal to 80 and less than 90. If Kathleen's grades on the first four tests were 75, 96, 97 and 87, solve for the range of grades x that she could score on the fifth test that would result in her getting a final grade of B . Write your answer as an inequality.
- 9 The cost of renting a bounce house includes a fixed charge of \$325 for setup and delivery, and a variable charge of \$175 for each hour the bounce house is rented for. Yvonne would like to rent a bounce house for a party, and is willing to spend a minimum amount of \$1200 and a maximum amount of \$1725. Solve for the range of hours t that she could hire the bounce house for. Write your answer as an inequality.
- 10 Ivan scored 88, 72, 77, 79 and 74 on his five tests during the semester. The average of those scores count for two-thirds of his final grade. He has a final exam at the end of the semester that will count for the other one-third. To get a B , his final grade must be greater than or equal to 80 and less than 90. Solve for the range of grades x that he could score on his final exam that would result in him getting a B . Write your answer as an inequality. Assume that 100 is the highest possible grade.



- 11** Determine whether the following values of x are solutions to the compound inequality.
 $2x < -3$ or $3x \geq 4$

a $x = 2$

b $x = 0$

- 12** Determine the solution for each of the following compound inequalities.

a $x < 4$ or $x > 5$

b $x > 2$ or $x > 4$

c $x > 2$ or $x < 5$.

d $x > 6$ or $x > 3$

- 13** For each of the following compound inequalities:

i Solve for x in the first inequality.

ii Solve for x in the second inequality.

iii Write the solution to the compound inequality.

iv Plot the solution set on a number line

a $x + 5 < -4$ or $x - 4 > 2$.

b $5x > -20$ or $-6x < 12$

c $\frac{x}{2} \leq 1$ or $7 \geq x - 3$

d $x - 5 < 6$ or $x + 5 > 6$

e $-5 + x > -4$ or $5x + 1 > 26$

f $2x - 18 < -0.4$ or $2x - 18 > 0.4$

Additional questions

- 14** For the following pairs of inequalities, plot the solution to the compound inequality that is Inequality 1 **and** Inequality 2 on a numberline.

a Inequality 1: $x \geq 2$

Inequality 2: $x \leq 7$

b Inequality 1: $x > 3$

Inequality 2: $x \leq 7$

- 15** For the following pairs of inequalities, plot the solution to the compound inequality that is Inequality 1 **or** Inequality 2 on a number line.

a Inequality 1: $x \leq 3$

Inequality 2: $x > 8$

b Inequality 1: $x < 8$

Inequality 2: $x > 3$

16 For the following compound inequalities:



- i Rewrite the compound inequality as a pair of inequalities joined by either **and** or **or**.
- ii Plot the solution to the compound inequality on a number line.
- iii Write the solution in interval notation.

a $-5 < x < 8$

b $-7 < x \leq 5$

17 For the following pairs of inequalities, state the solution to the compound inequality that is Inequality 1 **and** Inequality 2.

a Inequality 1: $-3x + 2 \leq -13$
Inequality 2: $-4x - 2 \geq 26$

b Inequality 1: $4x + 3 \geq -17$
Inequality 2: $3x - 4 < 5$

18 For the following pairs of inequalities, state the solution to the compound inequality that is Inequality 1 **or** Inequality 2.

a Inequality 1: $5x - 7 < -22$
Inequality 2: $-6x + 8 < -28$

b Inequality 1: $-6x + 8 < 26$
Inequality 2: $5x - 4 \leq 26$

19 Carlton tried to solve the following compound inequality, but made a mistake in his work:

Step 0: $-8 \leq 10 - 3x < 4$

Step 1: $-18 \leq -3x < -6$

Step 2: $6 \leq 3x < 18$

Step 3: $2 \leq x < 6$

Determine which step is incorrect and explain the error.

20 Skyler tried to plot the solution to the compound inequality $4x - 1 \leq 11$ and $\frac{x+1}{2} > 5$ on a number line, however his answer is incorrect.



Identify the error(s) and explain how to rectify them.